NERF

NeRF: Neural Radiance Fields

Mildenhall et al. | arXiv:2003.08934v2

Represent one static scene as a continuous neural 5-D radiance field and render novel views by differentiable volumetric integration along camera rays.

novel views · radiance field · volume rendering

1 Problem

- View synthesis must recover complex geometry, occlusion and view-dependent appearance from a limited set of posed images.
- Discrete voxels waste memory; earlier neural rendering often blurs fine detail or needs special scene representations.

2 Scene function

A CONTINUOUS 5-D FIELD

- An MLP maps spatial position (x,y,z) and viewing direction to volume density sigma and emitted RGB radiance.
- Density depends only on position; color also depends on direction, allowing non-Lambertian highlights.

3 Rendering

INTEGRATE ALONG EACH RAY

- Camera rays are sampled at 3-D points; alpha compositing approximates the classic volume-rendering integral.
- Because rendering is differentiable, pixel errors backpropagate through weights, colors, densities and sample locations.

4 Key design

- Fourier positional encoding lets the MLP represent high-frequency geometry and texture instead of spectral smoothing.
- A coarse network allocates a second set of samples to occupied regions, concentrating computation where it matters.

5 Evidence

- On synthetic objects and forward-facing real scenes, NeRF outperforms prior neural rendering baselines in PSNR, SSIM and LPIPS.
- Only RGB images and known camera poses supervise training; complex visibility and appearance emerge in the field.

6 Meaning & limits

A NEW 3-D REPRESENTATION

- NeRF unified implicit neural representations with differentiable rendering and reshaped 3-D vision and graphics.
- The original method trains one model per static scene, requires calibrated poses and is slow to optimize and render.